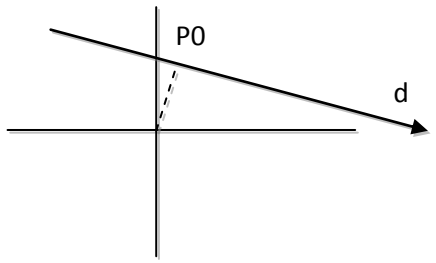
Line parametric equation:

$P(t) = P_0 + t(P_1 - P_0) = P_0 + t \cdot \vec{d}$

where the parameter *t* is a real number and d=P1-P0 is a line direction vector.

Using this representation P(0)=P0, P(1)=P1, and when $0 \le t \le 1$, P(t) is a point on the finite segment P0P1.

In my implementation I set P0 as closer point, so it respect $P_0 \cdot \vec{d} = 0$



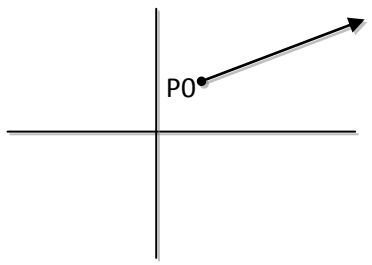
Line constructor:

Pass a generic point lies in line and a direction, to find P0 is sufficient substitute $P_0 = P_t - t \cdot \vec{d}$ in $P_0 \cdot \vec{d} = 0$ to obtain $(P_t - t\vec{d}) \cdot \vec{d} = P_t \cdot \vec{d} - t(\vec{d} \cdot \vec{d}) = P_t \cdot \vec{d} - t = 0$. Using $t_0 = P_t \cdot \vec{d}$ in the line equation you can find the P0 with $P_0 = P_t - t_0 \cdot \vec{d}$

```
public Line(Vector3 point, Vector3 dir)
{
    float t = Vector3.Dot(point, dir);
    d = dir;
    p = point - t * dir;
}
```

Ray parametric equation:

$P(t) = P_0 + t \cdot \vec{d}$



Using this representation, all P(t) are in $t \ge 0$, if *t* is negative P(t) are in negative direction

Segment parametric equation:

$$P(t) = P_1 f + P_0(1 - f) = P_0 + (P_1 - P_0) f$$

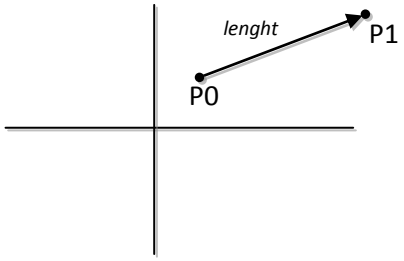
$$0 \le f \le 1$$

But to use a same formula of line I store origin point P0, direction vector and length of P1-P0. The first formula are used for linear interpolation function *lerp(f)*.

$$P(t) = P_0 + t \cdot \vec{d}$$

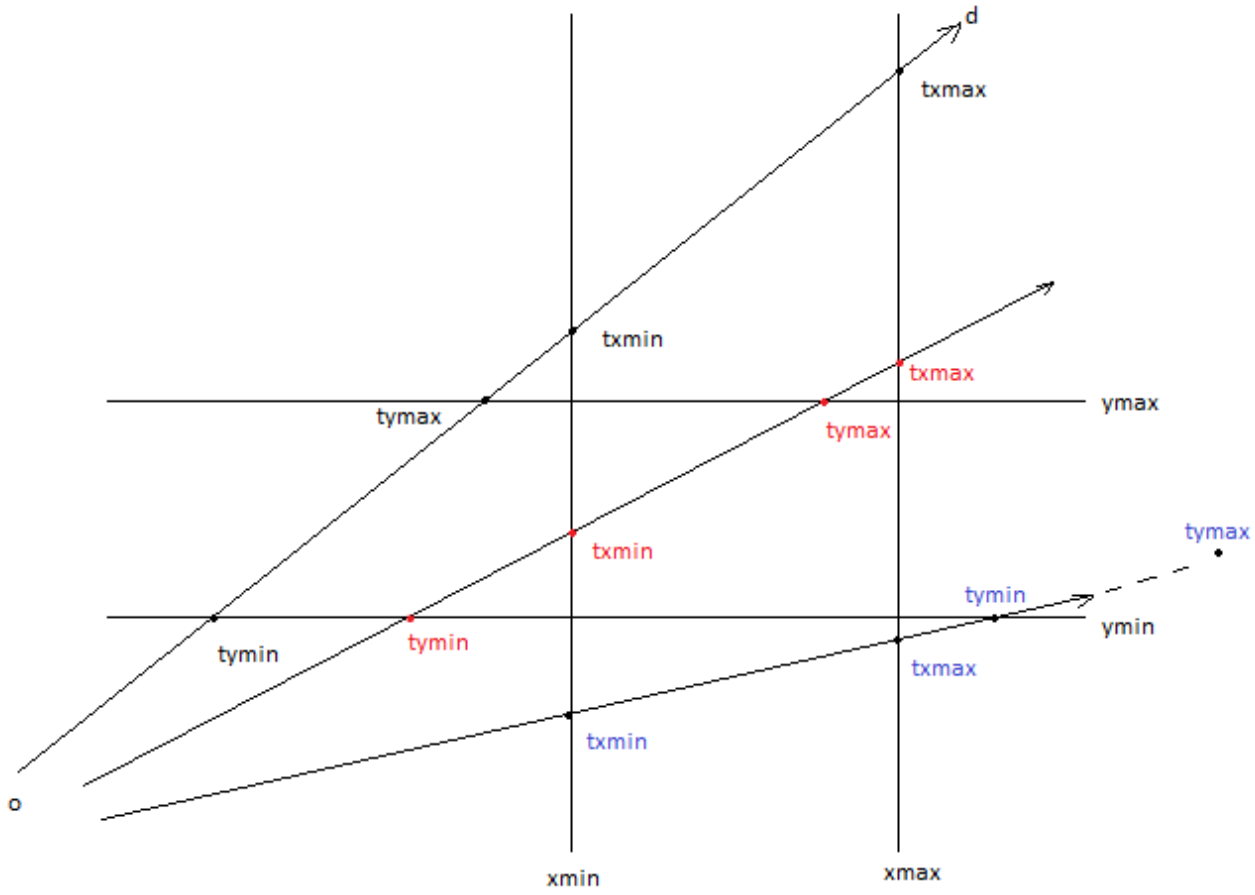
$$t = f * lenght$$

$$0 \le t \le lenght$$



Line – AABB intersection:

The simple 2d case :



Get a box defined with Max(xmax,ymax) and Min(xmin,ymin), the bounds of the area define a set of lines parallel to each axis of the coordinate. To find the parameter t where ray intersect each lines we use this formula, same for all 4 intersection

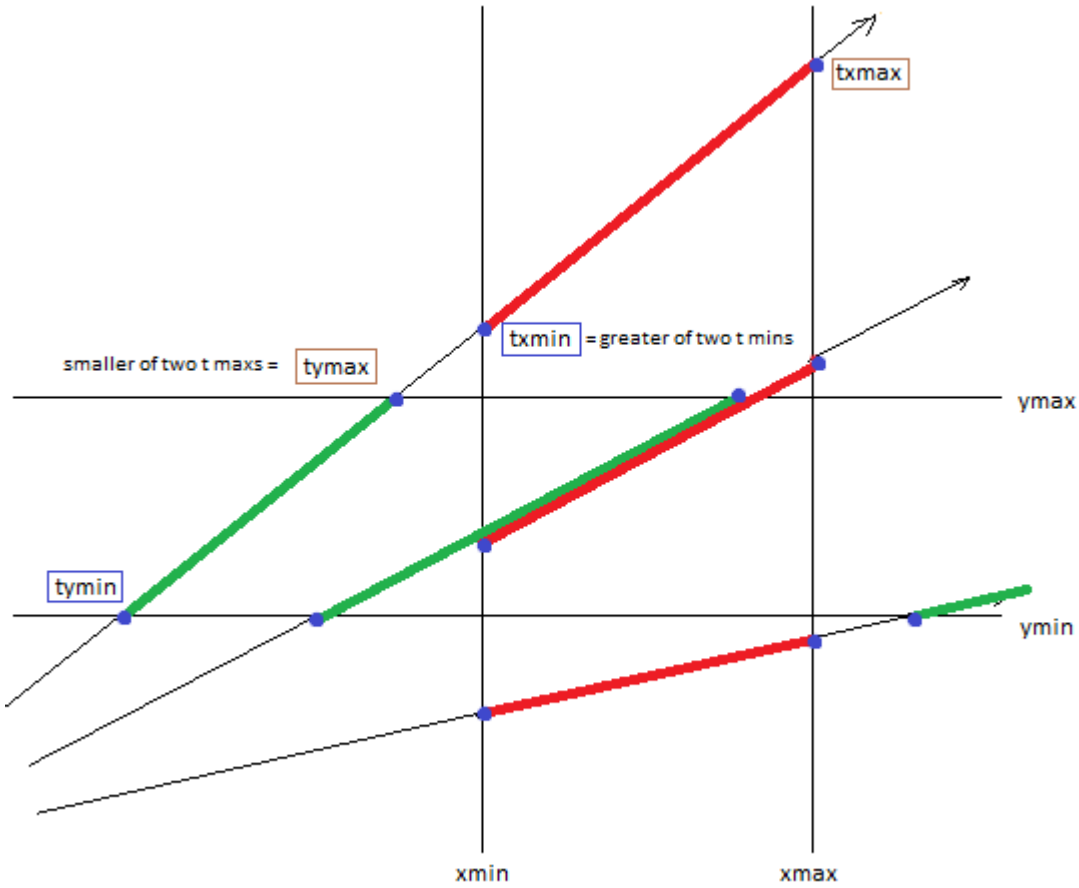
$$P(t) = P_0 + t \cdot \vec{d}$$

$$P(t_{x\min}) = x\min = P_{0x} + t_{x\min}\vec{d}_x$$

$$t_{x\min} = \frac{x\min - P_{0x}}{\vec{d}_x}$$

The line miss the box only in 2 case, when **txmax < tymin** or **tymax < txmin**

To explain well, we consider entering and leaving intersections separately , the ray intersect only when last entering is before first leaving:



So the formula of intersection extended for 3D is :

$$\boxed{MAX\left(t_{x\min}, t_{y\min}, t_{z\min}\right) < MIN\left(t_{x\max}, t_{y\max}, t_{z\max}\right)}$$